R-E selected: boundaries, recovery overlap and the next choice

15 October 2026 — analytical review; no traffic time stepping

Status. The user selects R-E, closing SEP-R1 at source-attribution scope. Ordinary A departures do not add independent recovery profiles. S-E-P remains selected. Exact work, marks, overlap and global event proofs remain open. This continuation stops at **RE-O1**, Type 2 G3: additive burdens O-S versus recent-mass renewal O-R. Neither overlap law is selected.

1. Retained interface and the trial isolated profile

Retain H1's entrance histories F_j, advice $v \in [v_-, v_+]$, finite terminal length b, initially empty histories, and

$$P_j(t) = F_j(t - b/v), \quad q_j = P_j - Y_j, \quad N_j = F_j - Y_j, \quad \lambda_j = \dot{P}_j, \quad g_j = \dot{Y}_j. \tag{1}$$

The B lane remains independent and unqueued; $B(t)$ below is instead the shared A/E budget. Write $K = wk_j$ and use

$$c(u) = \frac{Ku}{w+u}, \quad C(v) = c(v), \quad U(f) = \frac{wf}{K-f}, \quad 0 < R \le C(v_-), \quad u_R = U(R). \tag{2}$$

The trial node has ordinary cost one, $c_0 = 1 + \delta$, and

$$c_E = c_0 + \beta \frac{g_A}{C(v)} \frac{[u_A - u_E]_+}{v}, \quad g_A + c_E g_E \le B, \quad \eta = c_0 - \beta(1 + v_+/w) > 0. \tag{3}$$

Here $\delta, \beta \ge 0$. The sufficient η condition and representative trace marks $u_j = v$ when free and $u_j = U(g_j)$ when queued remain trial assumptions. The earlier note proves selected-regime roots, not a complete hybrid law. S-E-P protects a free A offer that fits B; queued A uses the equal-work/unused-work redistribution template, even when E is free.

For $a_r > 0$, the proposed isolated recovery profile is

$$\begin{aligned} V(\theta; u) &= \sqrt{(w+u)^2 + 2wa_r\theta} - w, \\ D(\theta; u) &= [R - c(V(\theta; u))]_+, \qquad \theta \ge 0. \end{aligned} \tag{4}$$

For $u < u_R$, its duration and integrated loss are

$$\Theta(u) = \frac{u_R - u}{a_r} + \frac{u_R^2 - u^2}{2a_r w}, \quad W(u) = \frac{k_c(u_R)(u_R - u)^2}{2a_r}, \quad k_c(z) = \frac{K}{w+z}. \tag{5}$$

For $u \ge u_R$ both are zero. Put $\Theta_{\max} = \Theta(0)$, $W_0 = W(0)$. Then $0 \le D \le R$, D decreases with age, $|D_\theta| \le a_r k_j$, and $|D_u| \le k_j$. The age includes backward propagation, not a numerical time step.

2. What the additive R-E law now proves

O-S is the existing trial overlap family, with physical reference mass $m = m_{\mathrm{ref}} = 1$ vehicle-equivalent:

$$p_S(t) = \frac{1}{m}\int_0^t D(t-s; u_E(s))g_E(s)ds, \qquad B_S(t) = [R - p_S(t)]_+. \tag{6}$$

This is one account. It does not create a vehicle queue or another constraint for each past passage. The initial account is neutral.

Strictly positive finite-time service, conditional on an admissible causal solution. Since $c_E \geq c_0$, work feasibility gives $g_E \leq B_S/c_0$. Differentiating the nonincreasing profiles gives, almost everywhere,

$$\dot{p}_S = \frac{g_E(t)D(0; u_E(t))}{m} + \frac{1}{m}\int_0^t D_\theta(t-s; u_E(s))g_E(s)ds \leq \frac{R}{m}g_E \leq \frac{R}{mc_0}B_S. \tag{7}$$

Before a hypothetical first zero of B_S, $B_S = R - p_S$ and

$$\dot{B}_S \geq -\frac{R}{mc_0}B_S, \qquad B_S(t) \geq R\exp[-Rt/(mc_0)] > 0. \tag{8}$$

Continuity rules out a finite first zero. This derives a bound; it does not insert a positive floor. An arbitrary externally prescribed admission history can exhaust (6); it need not be admissible in the coupled node. Neither (8) nor the earlier fixed-mode contraction proves global queue-event existence, uniqueness or empirical fidelity.

A reachable first loss, not just a possible source. At the retained H1 fast-front example, translated so first eligibility is $t = 0$,

$$R = 39/20, \quad \lambda_A = 1047/880, \quad \lambda_E = 125/176, \quad \delta = \beta = 1/10, \quad C = 3. \tag{9}$$

The earlier strict-onset calculation yields free A, queued E and

$$g_* = \frac{16716 - 183\sqrt{5234}}{5189}, \quad \frac{2230}{3343} < g_* < \frac{669}{968} < \lambda_E < R. \tag{10}$$

Thus $u_E = U(g_*) < u_R$ and, with empty history,

$$\dot{p}_S(0+) = \frac{g_*(R-g_*)}{m} > 0, \qquad \dot{B}_S(0+) = -\frac{g_*(R-g_*)}{m} < 0. \tag{11}$$

The local fixed-mode causal construction and strict onset margin justify this initial right-hand effect. These numbers are inherited diagnostic values, not selected task calibration or a positive-control parameter set. No complete A-delay witness, preferred advice or interior optimum follows. The O-R candidate below agrees until the first reference mass has passed.

3. Strict boundary compatibility and a benign full-task reduction

Write $a = \lambda_A$, $e = \lambda_E$. With queued E, its unique trial service solves $H_E(g_E) = B - g_A$ and satisfies $g_E \le (B - g_A)/c_0$. With both queues active, $g_A = B/2$. With free E and queued A, E's entire offer fits if $c_0 e \le B/2$; then $g_A = B - c_0 e$. These are current-state rules, not allocation latches.

Strict event	Condition before switch	Compatible right-hand behavior
E onset; A free	$a \le B$, $e > (B-a)/c_0$	Queued E has $g_E \le (B-a)/c_0 < e$, so its newborn queue grows.
A onset	$a > B$	Queued allocation has $g_A \le B < a$; if free E also fails its share, its queued root is below that free share.
E onset; A queued	$e > B/(2c_0)$	Queued E has $g_E \le B/(2c_0) < e$.
E clears; A free	$e < g_E^Q \le (B-a)/c_0$	Both free offers fit $a + c_0 e < B$.
E clears; A queued	$e < g_E^Q \le B/(2c_0)$	Free E gets e; A gets $B - c_0 e \ge B/2$. Cooperation persists.
A clears; E queued	$a < g_A^Q = B/2$	Free A's offer fits B; E is recomputed against its residual.
A clears; E free	$a < g_A^Q = B - c_0 e$	Both free offers fit $a + c_0 e < B$.

For simultaneous strict clearance of both queues, $a < B/2$ and $e < g_E^Q \le B/(2c_0)$ imply free admission is feasible. These statements check switch directions. They do not establish every grazing or simultaneous birth case, an absence of Zeno accumulation, or full hybrid uniqueness. The earlier no-spontaneous-birth guard remains required. A step size, queue epsilon, sliding convention or extra hysteresis has not been added.

Exact benign reduction. For one H1 advice, translate its eligible pulse to $0 < t < \ell/v$. On that interval

$$a(t) + e(t) = s := v(e_0 + a_0) < R, \quad e(t) \text{ nonincreasing}, \quad a(t) + c_0 e(t) = s + \delta e(t). \tag{12}$$

If $s + \delta e(0+) \le R$, the free offers fit from the outset and only become easier to serve. Free E has $u_E = v \ge u_R$, so $D = 0$ and no new account loss arises. Causal free-mode retention therefore gives

$$q_A = q_E = 0, \quad p = 0, \quad B = R, \quad J(v) = M(L - \ell/2)/v. \tag{13}$$

If this holds for every admitted advice, all observers have the fastest optimum. This is a declared exact negative case for either overlap candidate, not a reason to force queueing or choose parameters for a positive result.

4. A stronger simple rival and the overlap reference

Stationary-equivalent work. If g_E and its mark u_E have been constant over a full profile window, O-S gives

$$p_S = \frac{g_E W(u_E)}{m}, \qquad g_A + \left(c_E + \frac{W(u_E)}{m}\right) g_E = R \quad \text{at saturated service.} \tag{14}$$

A state-dependent instantaneous work relation therefore matches this stationary consequence. This is our deduction, not a new native invariant junction. Its transient selection and feasibility need their own contract. It is a serious simpler rival: a history benefit must concern consequential transients, not merely lower average discharge. Proper aggregate/history, scalar-statistic, native lane-aware and equal-information search rivals stay.

Point-reference screen. The homogeneous point case in Leclercq et al. (2016), Section 2, equation (1) evaluates cumulative passing over the interval to the next insertion. For a fixed mark $u < u_R$ and fixed headway h_0, capping our isolated service at R gives the following own interval-mean deduction:

$$\overline{B}_{\text{point}} = R - \frac{W_{h_0}(u)}{h_0}, \qquad W_h(u) := \int_0^h D(\theta; u)\, d\theta. \tag{15}$$

For $z_h = \min\{u_R, V(h; u)\}$,

$$W_h(u) = \frac{k_c(u_R)}{2a_r}\{(u_R - u)^2 - (u_R - z_h)^2\}. \tag{16}$$

Full-profile addition under the corresponding constant mean admission $g_E = m/h_0$ instead gives $\overline{p}_S = W(u)/h_0$. Hence

$$\overline{p}_S - \frac{W_{h_0}(u)}{h_0} = \frac{1}{h_0}\int_{h_0}^{\Theta(u)} D(\theta; u)\, d\theta > 0 \quad (0 < h_0 < \Theta(u)). \tag{17}$$

When $h_0 \geq \Theta(u)$ the difference is zero. This compares an additive mean closure with a capped homogeneous interval reference. It is not an exact native transient law, a primary H1 trajectory, or a proof that the closed-loop O-S model exhausts its budget. Equation (8) prevents the latter at finite time in an admissible initially neutral solution.

An exact constitutive diagnostic uses $w = 2, k_j = 3, R = 2, u = 1/2, a_r = 1/2$, $m = h_0 = 1$. It gives $u_R = 1$, $\Theta = 11/8$, and

$$W = 1/2, \quad W_1 = 6\sqrt{33} - 34, \quad W - W_1 = 69/2 - 6\sqrt{33} > 0. \tag{18}$$

The mark and mean admissions are prescribed for this screen, not asserted to solve the trial node. No traffic is time stepped. Variable marks and interacting restrictions still require a physical overlap closure.

5. A concrete recent-mass renewal alternative

For O-R, let $T_E(\xi)$ be a generalized inverse of the continuous admitted history Y_E. Endpoint choices at plateaus have zero mass measure. Keep the same profile and define

$$p_R(t) = \frac{1}{m} \int_{[Y_E(t)-m]_+}^{Y_E(t)} D\big(t - T_E(\xi); u_E(T_E(\xi))\big) d\xi, \qquad B_R = [R - p_R]_+. \tag{19}$$

Equivalently integrate (6) only over past mass with $0 \le Y_E(t) - Y_E(s) < m$. This uses predicted continuous mass, not integer vehicles, packets, a second traffic queue or initial hidden information. Only still-active profiles within the most recent reference mass contribute. At first admission and until $Y_E = m$, O-R and O-S give the same functional. Their later closed-loop trajectories need not be ordered.

The physical interpretation is renewal of an average affected stream. An admitted portion contributes while replacing an equal amount of oldest profile weight after the window is full. The same $m = 1$ reference unit now also sets a physical renewal mass. Its role must be defended, not changed as a harmless numerical resolution or tuned for a desired control result.

Bounds that follow immediately. For any fixed admissible history,

$$0 \le p_R(t) \le p_S[Y_E, u_E](t), \qquad p_R(t) \le R \min\{Y_E(t)/m, 1\}. \tag{20}$$

The comparison uses the same supplied history; it does not compare two endogenously different solutions. After last E admission, the retained window is fixed and its profiles decrease, expiring within $\Theta_{\max}$. A later fast E portion has zero new profile but can retire older slow weight. That is part of this renewal model and a material risk: it can release service too quickly when an older leader still controls it.

For fixed $g_E > 0$, mark u, and enough previous admissions,

$$h_0 = m/g_E, \qquad p_R = \frac{g_E}{m} \int_0^{h_0} D(\theta; u) d\theta = \frac{W_{h_0}(u)}{h_0}. \tag{21}$$

Thus O-R reproduces (15)'s mean loss in that restricted reference. Equation (19) is our transient extension, not the source's native algorithm. Its window cutoff needs its own causal/event qualification. The O-S inverse and fixed-mode history proof do not automatically establish O-R uniqueness.

Both candidates preserve one shared output and E-only new disturbance. O-S preserves every older physical aging tail but assumes independent addition. O-R limits repeated accounting but assumes recent-mass renewal adequately represents old restrictions. No new spatial field, longitudinal zone, approach-speed ODE or A-source penalty is introduced by either option.

6. Joined accounting, proof limits and stopping gate

For either overlap law, let $\overline{M}_E = P_E(L/v)$ and assume an admissible causal solution. Nonnegative kernels and (20) give

$$\int_0^\infty p(t)dt \le \frac{\overline{M}_E W_0}{m}, \qquad R - B = \min\{R, p\} \le p. \tag{22}$$

Using the prior work-conserving trial node, $c_{\max} = c_0 + \beta$, and last eligibility $t_* = L/v$, the existing clearance argument gives

$$T_{\text{clear}} \le t_* + \frac{c_{\max}(q_A(t_*) + q_E(t_*)) + \overline{M}_E W_0/m}{R}. \tag{23}$$

It uses an upper bound on available E mass before clearance, not an assumed successful final exit. This is a conditional bound, not proof of a full hybrid solution. No positive service floor is added.

For $M < k_j b$ and $\dot{F}_j \le C(v)$, the earlier finite-terminal vacancy proof is unchanged for nonnegative departures. Physical population and the complete local objective remain

$$s(t)\rho_0 + \sum_j N_j + \sum_j Y_j = M, \qquad J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E)dt. \tag{24}$$

Early-merged mass belongs to A, not a second forced E merge. A retired recovery profile is not deleted traffic. Post-line recovery memory is not post-line vehicle-time. The independent B lane has no waiting in this domain.

State and evidence. Current allocation uses current traffic and budget. Future recovery generally needs admission/speed history, not just the scalar reading. Under initial symmetry and known empty memory, total mass plus initial ending mass still reconstructs H1; unique reconstruction of the joined dynamics remains conditional on a qualified whole law. No full-lane-state necessity, new action split, gradient correctness or independent gradient usefulness is proved. The constant-A protection result remains; changing H1 arrivals need a consequential complete A-delay/MP6 witness. A positive local loss derivative in (11) does not supply that.

Source and scope. Carey's Section 5 own-lane/work ingredients and Jin's boundary consistency warning were rechecked at prior reading scope; neither supplies this trial node's full event proof. S13, the closed urban task, the closed ramp line and the model–SUMO S-P–SUMO S-L ladder retain their status. No reference simulator, fitting or numerical gradients ran.

Gate RE-O1. Select independent additive fading burdens O-S, or recent-mass renewal O-R. R-E source attribution stays closed and S-E-P stays selected. Provisionally qualify O-R first for its explicit overlap treatment and restricted point-reference correspondence, retaining O-S as a serious phenomenological rival. Both are concrete formulas with different physical assumptions; neither is selected. Stop before adopting the new overlap law. The approved joined pass and Step 3 remain in progress; Step 4 is unopened.